Analysis

First baseline run with representative subset

Results found in 0-rep-test
Takeaways:

- Best performer: Method 2
- Overall very dominant method, which was able to be faster than method 1 and get a much better MOTA than method 3
- Surprisingly poor performance from method 3, which was able to cut FNs to almost halve (22,299 vs 12,493), but it produced way too many FPs, which led to a very poor MOTA, which ended up being negative for some sequences (per_sequence_metrics.csv). Method 3 had better IDS, which reflects better tracking from BoTSORT.

Interpretation:

- Yolo26 + ByteTrack improves tracking (252 vs 298) with respect to Yolo11 + SORT which is due to ByteTrack being more open to low confidence detections to follow up on existing tracks.
- Method 3 produced too many FPs, therefore confidence value should be increased. Probably the fault lies with the detector, the tracker managed to improve IDS significantly.

Early hyperparam tuning

Results found in 1-overnight
Takeaways:

- tuning_m3_summary.csv
    - for RT-DETR a value of 0.55 for the confidence produces better results than the previous values of 0.25 and 0.45 tested. With respect to 0.45, the following is noted:
        - FP: 1364 vs 2566
        - FN: 5800 vs 4839
    - Therefore, the FPs reduction outweighs the FN increase, and the IDS improves from 41 to 24
    - This leads to a better MOTA, which was 21.42% for the full dataset
- tuning_m2_summary.csv
    - For ByteTrack, lower confidence produced better performance. The attempts to increase to 0.25, 0.35 and 0.45 from the original 0.2 confidence produced worst results.
    - In this case, FPs were higher but the reduction in FNs made up for it

- Important to note that IDS was actually higher in 0.2 conf than with 0.45 or 0.35, but IDF1 was still higher, primarily because recall was higher than the other values
- sahi_summary.csv
    - SAHI performed very poorly and the reason becomes clear while analyzing [final_summary_by_method.csv](). It is able to reduce the number of FNs, therefore more objects are being correctly identified, but it has the highest IDS, which means the tracker is getting confused because there are too many FPs (about 4 times as many as in methods 2 and 3)

Final hyperparam tuning & ablation

Results found in 2-tuning
Takeaways:

- Best method final: RT-DETR(conf=0.55)+BoT-SORT+GMC
    - GMC improves MOTA
        - Reduced IDS
    - conf=0.55 produces more FNs but reduces the FPs
- Best IDF1 is actually RT-DETR(conf=0.5)+BoT-SORT
    - Less precision but more recall compared to conf=0.55
    - Since recall is overall significantly lower than precision (for tested methods), it ends up bumping F1-score up by bringing recall closer to precision
- Best real-time: Yolo26+BoT-SORT: FPS 53.96
- gmc_ablation_summary.csv
    - Adding GMC always improved both MOTA (~1%) and IDF1 (~4%), proving that camera movement from the drone was one of the elements proving challenging for the tracker
    - Just relying on geometry didn't work as well because of this camera movement too
- `Uav0000182_00000_v` : only sequence where GMC ablation didn't produce a favorable result (off was better than on) [mot_diagnostics_by_sequence.csv]
- `Uav0000305_00000_v` : quite challenging for detectors, RT-DETR almost doubled recall compared with Yolo26, but it was still pretty low (31.4% vs 16%). On the other hand, its precision was 89%.
- SAHI ended up being counterproductive: more detection, but the increase in FPs outweighed the improvement in detection and tracking. Many false tracks were created
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